

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project | Python · PostgreSQL · Power BI | Antony | 2025

1. Business Problem Statement

A leading retail company is experiencing shifts in customer purchasing patterns across demographics, product categories, and shopping channels. The company wants to leverage its transactional data to better understand what drives customer decisions — including the influence of discounts, product reviews, seasonal trends, and payment preferences.

Business Question: *How can the company use customer shopping data to identify key trends, improve customer engagement, and optimise marketing and product strategies?*

2. Dataset Overview

Feature	Details
Total Records	3,900 customer transactions
Total Columns	18 original features
Customer Age Range	18 to 70 years old
Purchase Amount Range	\$20 to \$100 USD
Average Purchase Amount	\$59.76 USD
Average Review Rating	3.75 out of 5.0
Product Categories	Clothing, Accessories, Footwear, Outerwear
Missing Data	37 null values in Review Rating — handled via category median imputation

3. Data Cleaning & Preparation (Python)

The dataset was loaded and cleaned using Python (Pandas) before being loaded into PostgreSQL. Steps performed:

- Missing Value Handling: Found 37 missing values in Review Rating. Used category-wise median imputation rather than overall median for greater accuracy.
- Column Standardisation: All column names converted to lowercase with underscores (snake_case).
- Column Renaming: Renamed purchase_amount_(usd) to purchase_amount.
- Feature Engineering — Age Group: Created age_group using pd.qcut into 4 equal groups: Young Adult, Adult, Middle-aged, Senior.
- Feature Engineering — Purchase Frequency: Mapped text-based frequency to numeric days (Weekly=7, Monthly=30, Annually=365 etc.).
- Duplicate Column Removal: Verified discount_applied and promo_code_used were identical — dropped promo_code_used.
- Database Integration: Connected to PostgreSQL using SQLAlchemy and psycopg2 and loaded the cleaned DataFrame as the customer table.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 19 columns):
customer_id      3900 non-null int64
age              3900 non-null int64
gender           3900 non-null object
item_purchased   3900 non-null object
category         3900 non-null object
purchase_amount  3900 non-null int64
location         3900 non-null object
size            3900 non-null object
color           3900 non-null object
season          3900 non-null object
review_rating    3900 non-null float64
subscription_status 3900 non-null object
shipping_type    3900 non-null object
discount_applied 3900 non-null object
previous_purchases 3900 non-null int64
payment_method   3900 non-null object
frequency_of_purchases 3900 non-null object
age_group        3900 non-null category
purchase_frequency_days 3900 non-null int64
dtypes: category(1), float64(1), int64(5), object(12)
memory usage: 552.5+ KB

```

4. Business Analysis Using SQL (PostgreSQL)

Ten business queries were written in PostgreSQL to answer key questions about customer behaviour, product performance, and revenue drivers:

Q1. Revenue by Gender

Male customers generated \$157,890 vs Female customers \$75,191. Males account for 68% of total revenue — the primary target segment for premium campaigns.

	gender text 	revenue numeric 
1	Female	75191
2	Male	157890

Q2. High-Spending Discount Users

839 customers used a discount but still spent above the average of \$59.76. These are high-value targets for upsell campaigns.

Q3. Top 5 Products by Review Rating

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78). These highly rated products should be featured in marketing.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Q4. Shipping Type vs. Purchase Amount

Express (\$60.48) vs Standard (\$58.46) — minimal difference. Shipping type does not significantly influence spend.

Q5. Subscriber vs. Non-Subscriber Spend

Non-subscribers: 2,847 customers, \$170,436 revenue. Subscribers: 1,053 customers, \$62,645 revenue. Average spend is nearly identical — subscription status does not affect spend but the volume gap shows a major conversion opportunity.

Q6. Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%) have the highest discount usage. These products may be over-reliant on discounts.

Q7. Customer Segmentation

Loyal customers (10+ purchases): 3,116. Returning (2-10): 701. New (1 purchase): 83. Strong retention but very limited new customer acquisition.

	customer_segment text	total_customers bigint	total_purchases numeric	avg_purchases numeric
1	Loyal	3116	94714	30.40
2	Returning	701	4074	5.81
3	New	83	83	1.00

Q8. Top 3 Products per Category

Accessories: Jewelry, Sunglasses, Belt | Clothing: Blouse, Pants, Shirt | Footwear: Sandals, Shoes, Sneakers | Outerwear: Jacket, Coat, Hoodie.

	category text	item_purchased text	total_orders bigint	item_rank bigint
1	Accessories	Jewelry	171	1
2	Accessories	Sunglasses	161	2
3	Accessories	Belt	161	3
4	Clothing	Undergarments	171	1
5	Clothing	Pants	171	2
6	Clothing	Shirt	169	3
7	Footwear	Sandals	160	1
8	Footwear	Shoes	150	2
9	Footwear	Sneakers	145	3
10	Outerwear	Jacket	163	1
11	Outerwear	Coat	161	2

Q9. Repeat Buyers & Subscriptions

Of customers with 5+ purchases: 2,518 are non-subscribers vs 958 subscribers. Most loyal repeat buyers have not subscribed — significant opportunity.

Q10. Revenue by Age Group

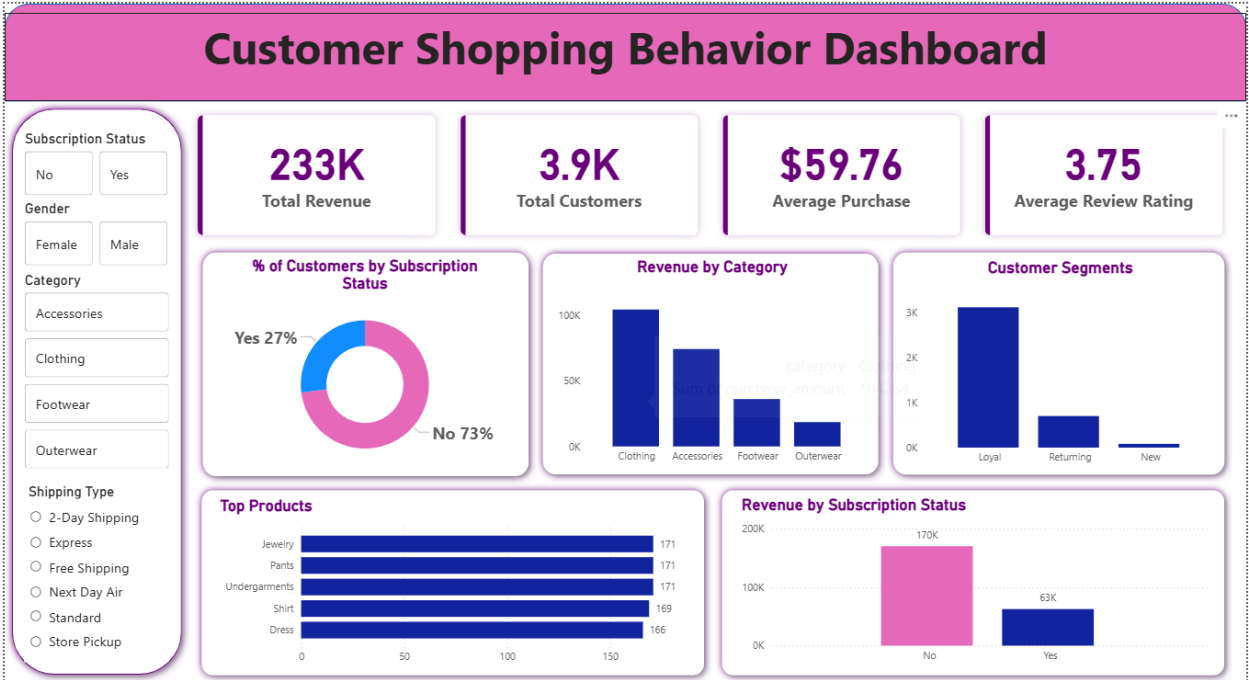
Young Adults: \$62,143 | Middle-aged: \$59,197 | Adult: \$55,978 | Senior: \$55,763. Revenue is balanced across all age groups with Young Adults slightly ahead.

	age_group text	revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Power BI Interactive Dashboard

An interactive dashboard was built in Power BI with a pink and purple colour theme. Key components:

- KPI Cards: Total Revenue (233K), Total Customers (3.9K), Average Purchase (\$59.76), Average Review Rating (3.75)
- Donut Chart: % of customers by subscription status (73% No, 27% Yes)
- Bar Chart: Revenue by Category — Clothing leads at ~\$103K
- Bar Chart: Customer Segments — Loyal, Returning, New
- Bar Chart: Revenue by Age Group — Young Adults lead
- Bar Chart: Sales by Age Group
- Slicers: Filter by Subscription Status, Gender, Category, Shipping Type



8. Conclusion

This project demonstrates a complete end-to-end data analytics workflow — from raw data ingestion and cleaning in Python, to structured business analysis in PostgreSQL, to interactive visualisation in Power BI. The analysis revealed strong customer loyalty but a significant subscription conversion gap, clear gender-based revenue dominance, and product-level insights that can directly inform marketing and inventory decisions.

The findings provide a clear foundation for the retail company to make data-driven decisions that improve customer engagement, optimise discount strategies, and drive sustainable revenue growth.